# Bitcoin Revived: Innovative Vitality and Maintenance of Decentralization

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**Abstract**

The emergence of Bitcoin has initiated people's exploration and practice in the field of decentralized electronic currency. It has provided valuable experience for humanity to seek new types of currencies more suitable for the future digital era and even to explore new ways of cooperation between people in the future. Meanwhile, observations reveal that the current operation mode of Bitcoin may face the risk of deviating from the decentralized development direction of Bitcoin. To address this, this paper proposes a Bitcoin revival scheme. On the basis of fully abiding by the Bitcoin whitepaper, the scheme adheres to a software evolution strategy that is always compatible with the early software versions of Bitcoin. The purpose is to fully explore and utilize the previously accumulated experience, ensure the system continues to develop in the direction of

decentralization, and provide better conditions for individual innovation.

**1. Direction**

The decentralized direction of future currencies has demonstrated strong vitality. Since the birth of Bitcoin, various digital currencies centered on the decentralized direction have emerged one after another. The price of Bitcoin has even highlighted the extent to which it is sought after by the market. In some regions, people even take the risk of violating laws to engage in such innovative activities—all of which demonstrate the strong vitality of the decentralized direction of future currencies. It can be predicted that the decentralized direction of currencies is becoming a golden key to unlocking the ideal digital era in the future. As the initiator and pioneer of the decentralized direction of currencies, Bitcoin should unswervingly adhere to long-term development along this direction.

**2. Risks**

The current Bitcoin’s potential incompatibility with early software versions may lead it to deviate from its original course and return to the old path of centralized currencies.

Bitcoin originated from Satoshi Nakamoto’s Bitcoin whitepaper, and Satoshi Nakamoto developed the early version of Bitcoin software based on this whitepaper. Undoubtedly, the early version of Bitcoin software is an operable currency system that best embodies the decentralized ideology in the Bitcoin whitepaper. However, in the subsequent software version updates of Bitcoin, compatibility with the early Satoshi Nakamoto software version has not been consistently maintained. After more than a decade of software updates, the early Satoshi Nakamoto version of Bitcoin software can no longer connect to the Bitcoin network normally because it is rejected by most mainstream version clients. This inevitably raises the question: Can a system that even fails to support the normal connection of the Bitcoin software developed by Satoshi Nakamoto himself still be regarded as a Bitcoin system? Our view is that a system incompatible with the original Bitcoin software may not embody the decentralized ideology in the Bitcoin whitepaper and thus cannot be considered a true Bitcoin system. This is because software needs to be continuously upgraded and updated; after a system becomes incompatible with the original Bitcoin software, each subsequent software upgrade may cause its development direction to deviate. As time goes by, this deviation will become increasingly significant, and eventually, there may be a risk of the system becoming unrecognizable.

### 3. Measures

The process of reorganizing the network using the early version of the software and consistently adhering to a software evolution strategy compatible with the early Bitcoin versions is called "Bitcoin Revival". The early version of Bitcoin software, derived from the Bitcoin whitepaper and developed by Satoshi Nakamoto himself, is the safest choice for embodying the decentralized direction of currency as outlined in the whitepaper. People can carry out compatible innovation based on this foundation. The more than a decade of development history and technological progress of the Bitcoin system have provided valuable experience and abundant resources for compatible innovation. There are two approaches to software compatibility with early Bitcoin versions: benign compatibility and malicious compatibility.

- **Benign compatibility**: It involves participating in reorganizing the network using the early version of the software, expanding software functions, providing various resources required for network operation, and contributing to the expansion of the network scale.

- **Malicious compatibility**: It refers to exploiting vulnerabilities in the early version of the software to launch attacks, hindering or disrupting the normal operation of the system. The positive significance of malicious compatibility lies in its ability to help expose system vulnerabilities, thereby facilitating rapid fixes and improvements.

## 4. Outlook

Anyone can participate in compatible innovation to display their talents and create the future. Due to the open-source nature of Bitcoin code and the popularization of AI technology, Bitcoin Revival has actually lowered the development threshold for Bitcoin software, making it more feasible for more individuals to participate in the innovation of Bitcoin software development. In the future, the Bitcoin consensus mechanism will likely shift from the original model of "community negotiation plus team development" to a development model featuring "more individual innovation plus individual implementation". Innovation should not be limited to negotiating a single possible approach for implementation; instead, there should be more infinite possibilities for practice. The compatible innovation method proposed by Bitcoin Revival makes the latter scenario possible.

**5. Protection**

Adopting the Pixiu Algorithm to break free from hash rate dependence. In the early stage of Bitcoin Revival, due to the relatively low overall hash rate of the entire network, there may be a risk that malicious attackers launch fork attacks on the entire network by deploying powerful hash rates. As a countermeasure, nodes need to incorporate the "only-in, no-out" Pixiu Algorithm for protection, proactively rejecting potential block depth reorganizations. This ensures that the system can operate safely without requiring significant hash rate investment, thereby breaking free from hash rate dependence, achieving energy conservation and environmental protection, and ensuring economic efficiency.

References

【1】Satoshi Nakamoto，《Bitcoin: A Peer-to-Peer Electronic Cash System》，2008